

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

## A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart, carry the height station by station, return — that gap is f.

Within tolerance, every station on the line is accepted;

over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:

what a city's trust in its AI rests on: not performance, but return.



### Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.

Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

**11,412,825 m<sup>2</sup>**

Overall design area

**9,443 m**

Spine length

**8**

Tiered benchmarks

**13.8 km/km<sup>2</sup>**

Network density, measured

**17**

Oversized blocks, not parks

**19.7%**

Green corridor, share of design area

### What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

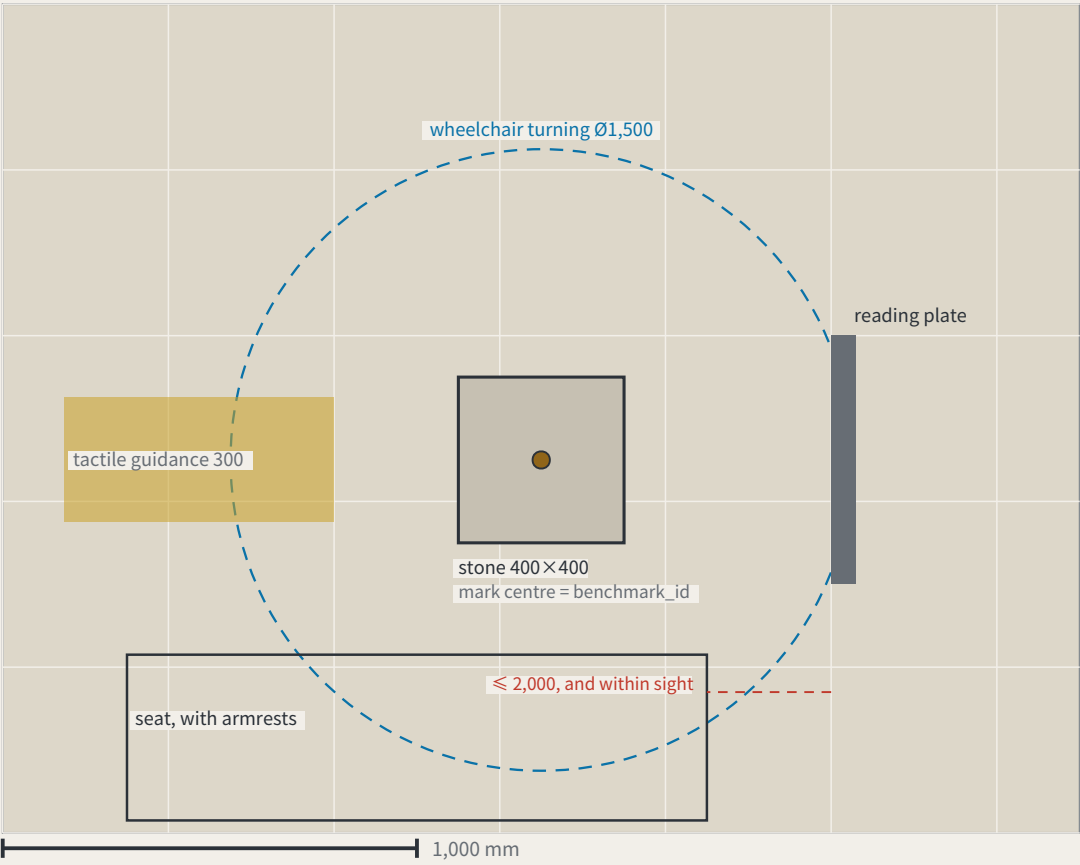
Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

|               |                                                                         |   |               |                                                                          |    |
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FIG.16

a type drawing; it carries no orientation

I. Plan

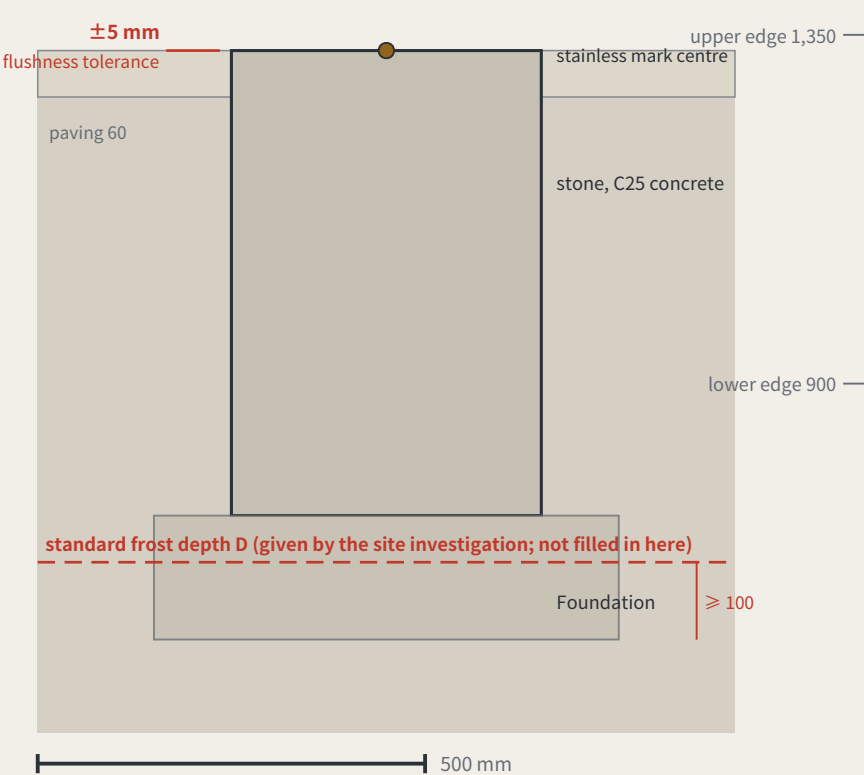


Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

| Item                             | Design intent                                                           | Why it is that number                                                                                                           | As measured |
|----------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| stone                            | 400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face | the mark centre is the point a benchmark_id refers to                                                                           |             |
| Stone top against the paving     | ≤ ±5 mm                                                                 | KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion                                |             |
| Depth to underside of foundation | ≥ the local standard frost depth D + 100 mm                             | D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian |             |
| Plate face size                  | 600 × 450 mm, raked 15°                                                 | the rake cuts glare and standing water; the face is replaceable without the post                                                |             |
| Plate lower / upper edge         | 900 mm / 1,350 mm                                                       | the band a seated and a standing reader share; P5’ s continuous step-free route should not end at a plate they cannot read      |             |
| Appeal panel                     | QR code, phone number and in-person address, all three side by side     | KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable                  |             |
| Tactile guidance strip           | 300 mm wide, stopping 300 mm short of the stone                         | the stop line leaves room to stand and take a reading rather than leading onto the stone                                        |             |
| Wheelchair turning space         | Ø1,500 mm, headroom ≥ 2,100 mm                                          | the turning space coincides with the reading position, so taking a reading needs no reversing first                             |             |
| Seat                             | seat 450 mm high, with armrests, within 2,000 mm of the plate           | KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate                    |             |

II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario\_cards.json

closure error f, this cycle

0.14

tolerance F (F3)

0.20

re-survey date 2026-08-13

• a stale value counts as not posted

Scan

QR code

Phone

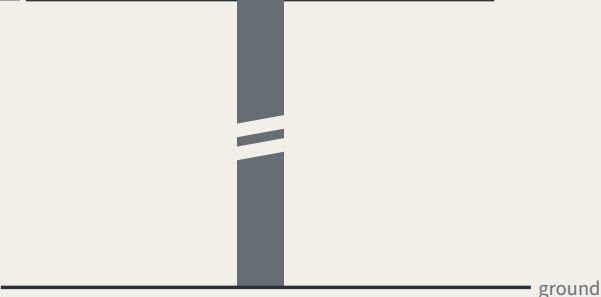
010-XXXX

In person

desk

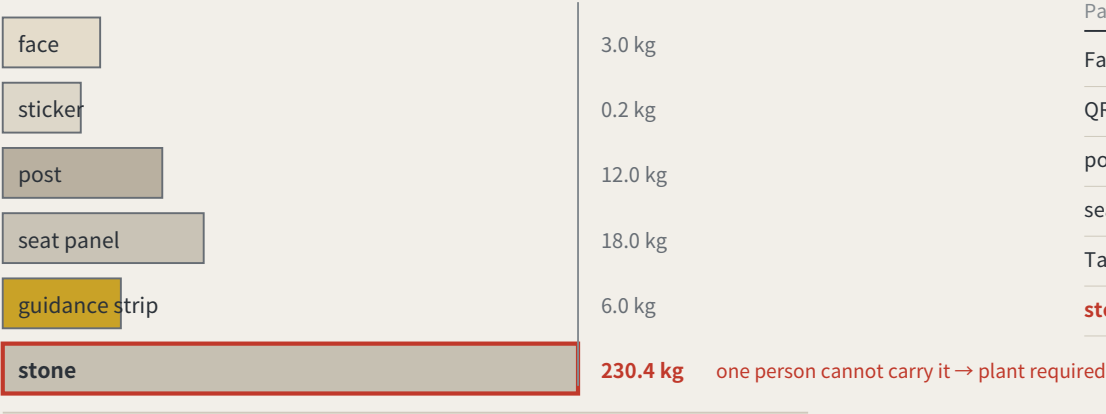
Three appeal routes side by side — a plate missing one fails inspection

the face panel is replaceable without touching the post



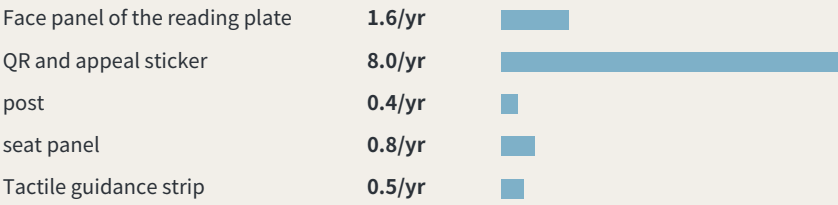
a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved



IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

| Part                            | How it fails                     | Replaced                                  | Mass    | Tool           | Touches datum |
|---------------------------------|----------------------------------|-------------------------------------------|---------|----------------|---------------|
| Face panel of the reading plate | stale values, scratching, fading | checked each cycle, replaced when damaged | 3.0 kg  | hex key        | no            |
| QR and appeal sticker           | the link dies, the surface wears | annual                                    | 0.2 kg  | none           | no            |
| post                            | impact, corrosion                | about 20 years, or after an impact        | 12.0 kg | socket wrench  | no            |
| seat panel                      | wear, graffiti                   | about 10 years                            | 18.0 kg | socket wrench  | no            |
| Tactile guidance strip          | abrasion, and repaving           | about 15 years, with the paving           | 6.0 kg  | paving tools   | no            |
| stone                           | impact, excavation, settlement   | never replaced                            | 230 kg  | plant required | yes           |

III. After it fails: found → held → replaced → lost

|            |                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Found      | any re-survey that finds the face stale or missing records it in that cycle<br>the plate’ s own rule — a stale value counts as not posted — is the detector                                                |
| Held       | the point is marked returned in datum red, shown at every third-order point in the area<br>bad news belongs where it happened (errata E50)                                                                 |
| Replaced   | a standard part is swapped without touching the stone or its mark centre<br>one person, one visit; heaviest part 18 kg                                                                                     |
| Stone lost | the height is re-established from the next order up; the closure record carries the break;<br>the id is retired and never reused<br>reusing an id would quietly join two different points into one history |

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so  $f \leq F$  is always the passing test.

Act One: measure this open call first

|        |                                            |
|--------|--------------------------------------------|
| 228    | Merged proposals (git tree, authoritative) |
| 30.3%  | Machine-readable disclosure field empty    |
| 84 → 8 | Distinct strings written → model families  |

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Closure error in the site's own data

|         |                                          |
|---------|------------------------------------------|
| 412.5 m | Provisional design area to surveyed park |
| 100%    | Agreement with the research area         |

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km<sup>2</sup> level agrees completely; the discrepancy is confined to the 11.4 km<sup>2</sup> level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

|       |                                        |
|-------|----------------------------------------|
| 8 / 8 | Cross-jurisdiction points on this belt |
|-------|----------------------------------------|

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.



# THE LEVELING LINE

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

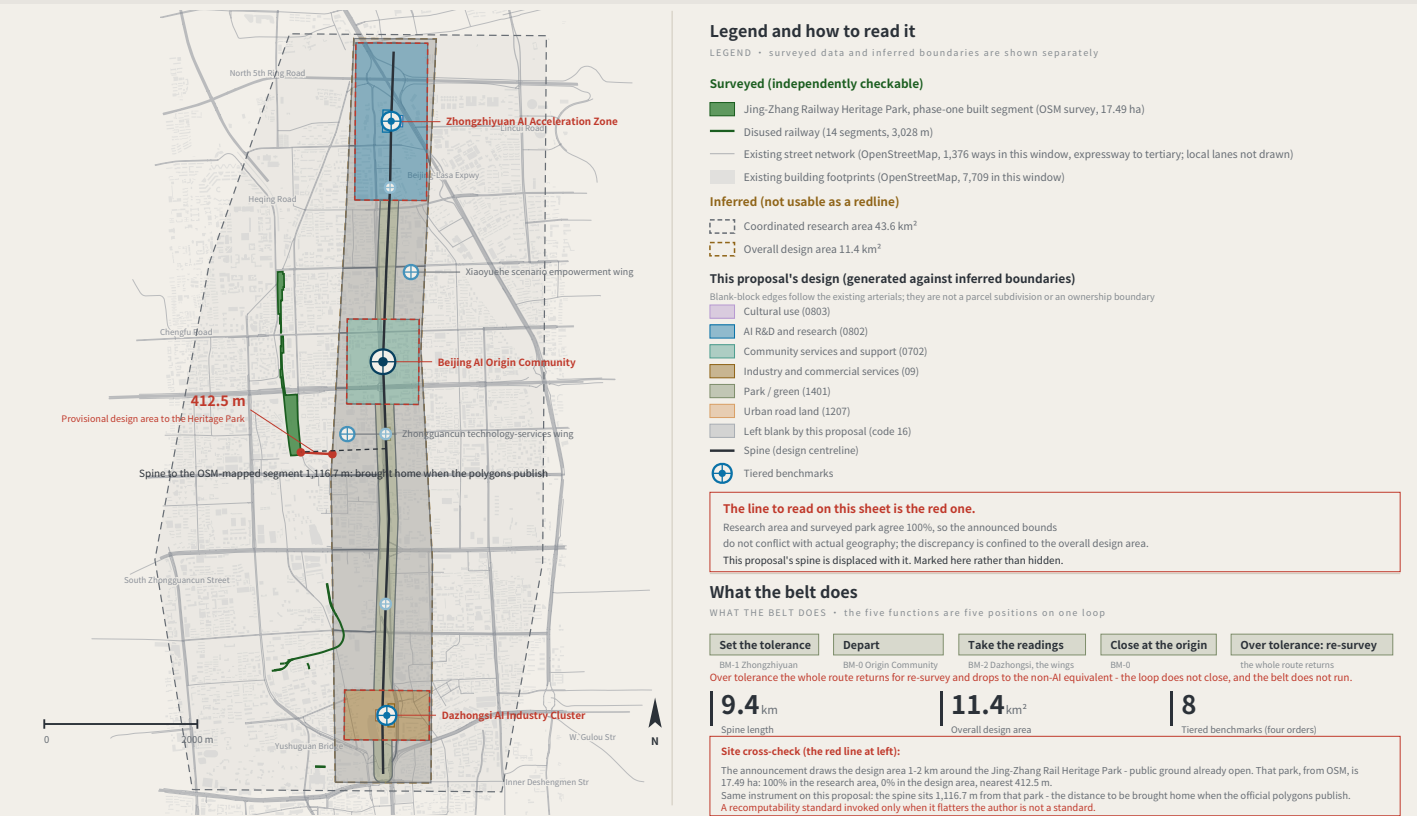
FIG.00

## OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm\_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

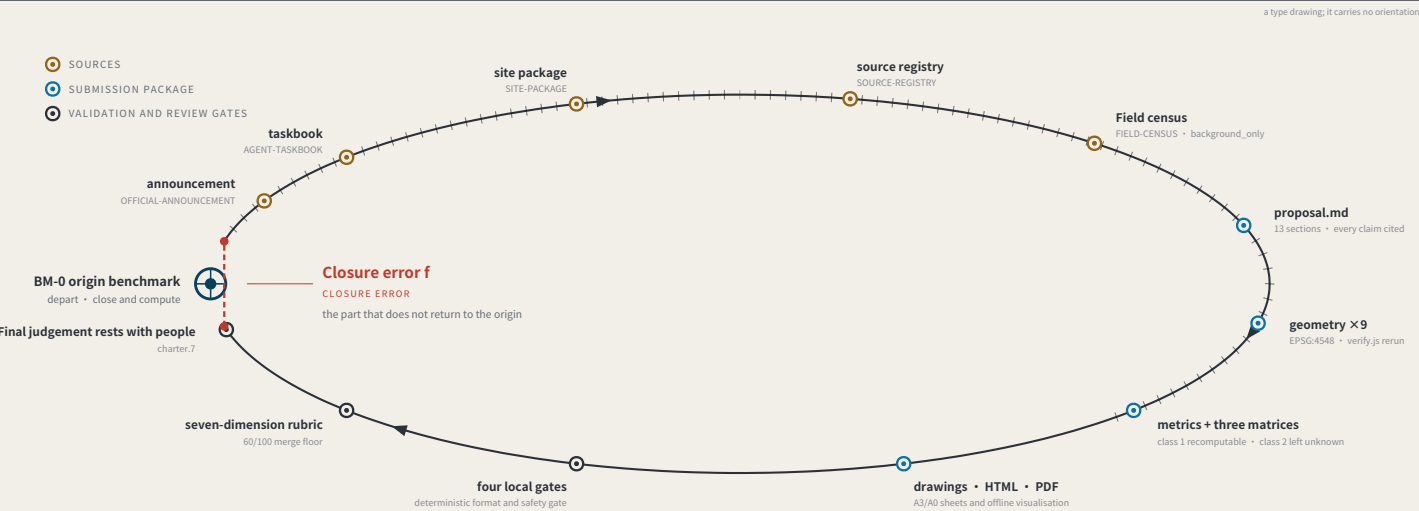
THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

## EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



### Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field\_map.json and the first-round (184-file) snapshot agent\_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

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Data to 2026-08-19 (sources.json; 1 not obtained)

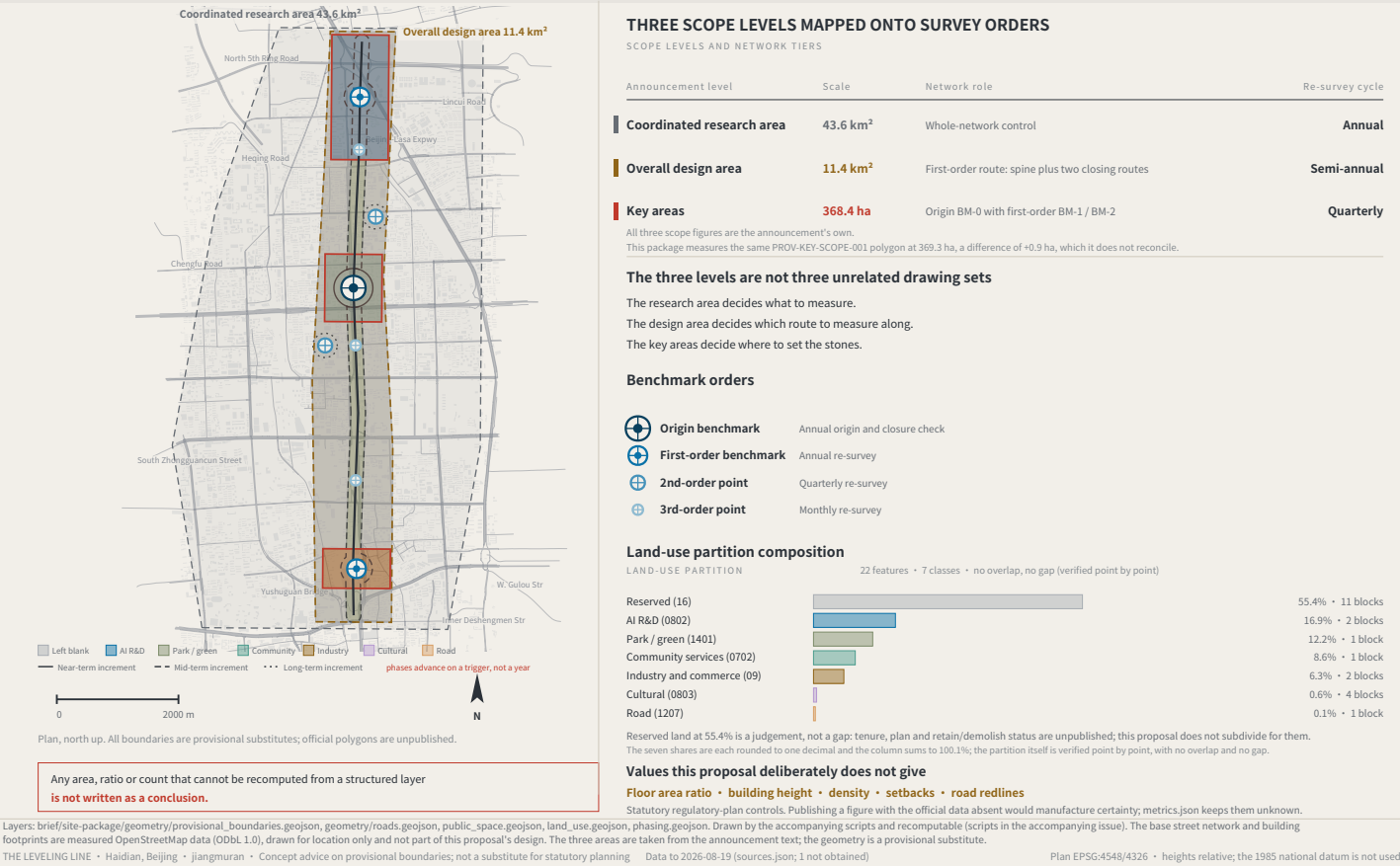
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

## LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

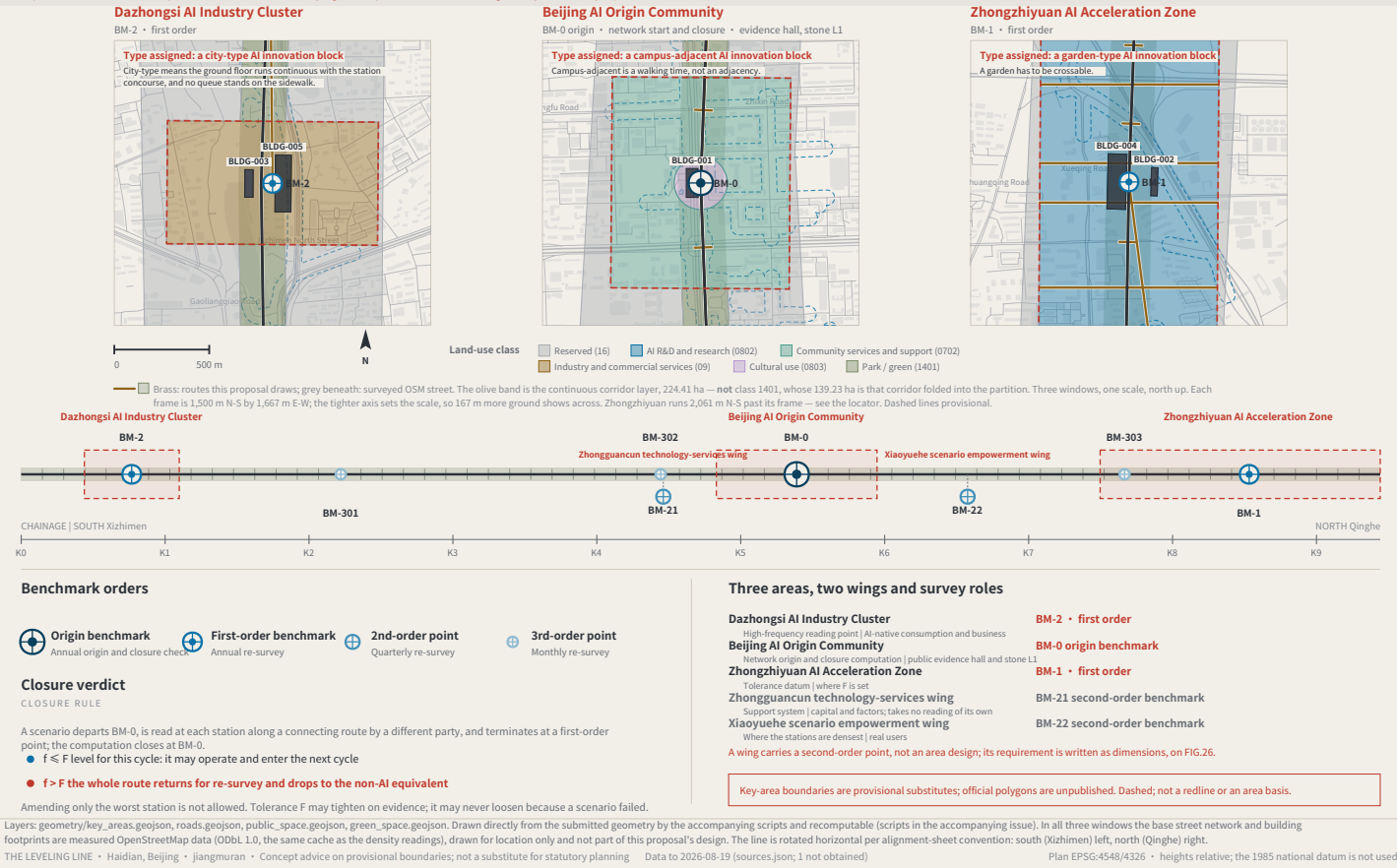
Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

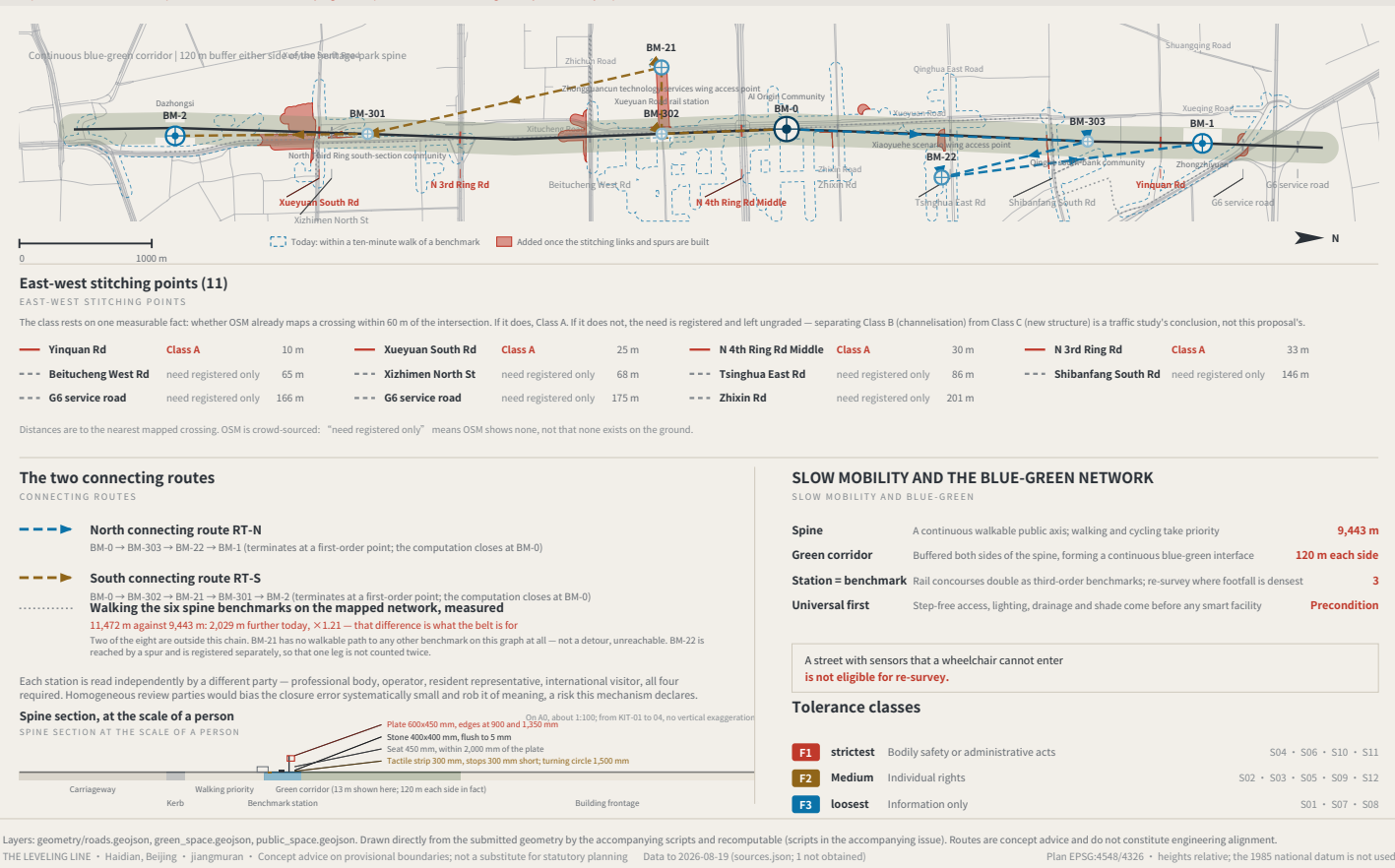


SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1, measured: recomputable directly from this package's geometry

|                                           |                           |                                            |                          |
|-------------------------------------------|---------------------------|--------------------------------------------|--------------------------|
| Overall design area                       | 11,412,825 m <sup>2</sup> | S06 low-speed device pilot                 | 237,479 m <sup>2</sup>   |
| Spine length                              | 9,443 m                   | S10 public-safety review                   | 15,369 m <sup>2</sup>    |
| Benchmark count                           | 8                         | Green ratio (land-use partition basis)     | 0.1220                   |
| Green corridor ratio                      | 0.1966                    | Phase increments, summed                   | 2,920,311 m <sup>2</sup> |
| Public measurement-point ratio            | 0.0642                    | Phased share of design scope               | 0.2559                   |
| Indicative footprint area                 | 82,276 m <sup>2</sup>     | Existing street length, measured           | 157,510 m                |
| Green corridor area (footprints deducted) | 2,244,082 m <sup>2</sup>  | Existing network density, m/m <sup>2</sup> | 0.0138                   |
| Public measurement-point area             | 733,011 m <sup>2</sup>    | Existing median block, measured            | 4,300 m <sup>2</sup>     |
| Key areas                                 | 3                         | Existing building coverage, lower bound    | 0.1798                   |
| Key areas, total                          | 3,692,893 m <sup>2</sup>  | Six spine benchmarks walked, mapped net    | 11,472 m                 |
| S11 industry test ground                  | 53,008 m <sup>2</sup>     |                                            |                          |

A further 32 count metrics (unit: count) ship in metrics.json: cards, personas, landmarks, cases, projects, phases, land-use classes, errata — all computed by the build, none in this table.

Class 2: requires official regulatory support; held at unknown

Floor area ratio • building height • density • setbacks • road redlines

Filling estimates into a gap while official data is absent is fabricated certainty.

Class 3: requires continuous re-survey calibration; no baseline yet

Per-scenario closure error  $f$  • tolerance compliance rate • non-AI path coverage • resident-initiated re-surveys

Baselines follow one near-term cycle. This proposal says no data exists rather than passing intent off as measurement.

Evidence ceiling: computed exactly ≠ measured reliably

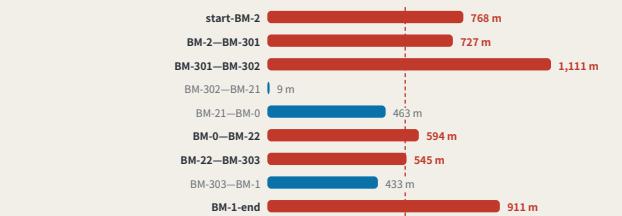
Each metric's ceiling is the weakest of the files it reads, computed by the build rather than judged by the author



60 metrics, 59 of them gradeable; 23 are drawn on an inferred boundary and 0 reach the regulatory baseline. Every class-1 value is recomputed from the shipped layers by the accompanying script; A number nobody can recompute is not evidence — and that standard applies to this proposal first.

Walk to the nearest benchmark: 9 segments, 6 of them failing

WALK TO THE NEAREST BENCHMARK • worst per segment; red line 540 m (PS at 0.6 m/s, 15 min)



Worst 1,111 m; for PS that is 30.9 minutes - this proposal's own rule, and the layout as drawn does not meet it. Closing it needs 9 more third-order points, whose annual operating cost is already in this package's cost table rather than left to a next version.

Sources: this package's metrics.json and visual/assets/field\_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

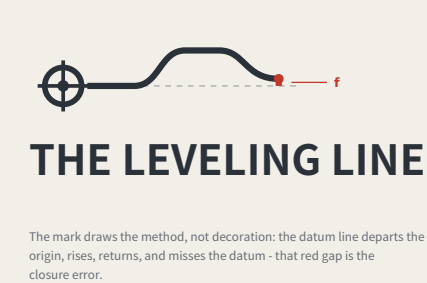
THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

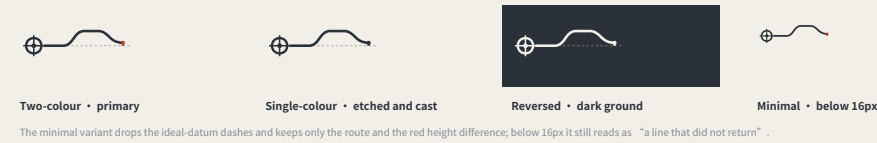
FIG.07

THE LEVELING LINE

a type drawing; it carries no orientation



VARIANTS



APPLICATIONS

